COMPUTER NETWORKS

SDN MININET PROJECT REPORT

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PROJECT TITLE - Static Routing using SDN with POX Controller in Mininet

1. Introduction

Software Defined Networking (SDN) is a modern networking paradigm that separates the control plane from the data plane, enabling centralized control of network behavior. In traditional networks, switches and routers make independent forwarding decisions, making management complex. In SDN, a centralized controller determines how packets should be handled, while switches simply execute these instructions.

This project demonstrates static routing using an SDN controller (POX) in a Mininet environment. The controller installs flow rules using the OpenFlow protocol to control packet forwarding and enforce network policies.

2. Objective

- To understand SDN architecture
- To implement static routing using POX
- To design match–action flow rules
- To demonstrate controller–switch interaction
- To analyze network behavior using ping, iperf, and Wireshark

3. System Architecture

- **Controller (POX):** Makes forwarding decisions
- **Switches (s1, s2, s3):** Execute flow rules
- **Hosts (h1, h2, h3):** Generate traffic

Communication is done using the OpenFlow protocol.

4. Network Topology

```
h1 — s1 — s2 — s3 — h3
          |
          h2
```

Screenshot 1: Mininet Topology

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s2-eth1
h3 h3-eth0:s3-eth1
s1 lo:  s1-eth1:h1-eth0 s1-eth2:s2-eth2
s2 lo:  s2-eth1:h2-eth0 s2-eth2:s1-eth2 s2-eth3:s3-eth2
s3 lo:  s3-eth1:h3-eth0 s3-eth2:s2-eth3
c0
```

5. Implementation (CODE – myrouter.py) – All 3 hosts can communicate

The controller handles PacketIn events and installs flow rules using match–action logic.

```
from pox.core import core
import pox.openflow.libopenflow_01 as of

log = core.getLogger()
```

```
//Packet In Handler
def _handle_PacketIn(event):
    packet = event.parsed

    if not packet.parsed:
        return

    dpid = event.connection.dpid
    in_port = event.port

    msg = of.ofp_flow_mod()
    msg.match = of.ofp_match.from_packet(packet)

    # SWITCH 1 (s1)
    if dpid == 1:
        out_port = 2 if in_port == 1 else 1

    # SWITCH 2 (s2) - FLOODING (FULL CONNECTIVITY)
    elif dpid == 2:
        if in_port == 1:
            out_port = of.OFPP_FLOOD
        elif in_port == 2:
            out_port = of.OFPP_FLOOD
        elif in_port == 3:
            out_port = of.OFPP_FLOOD
        else:
            return
```

```
    # SWITCH 3 (s3)
    elif dpid == 3:
        out_port = 2 if in_port == 1 else 1

    else:
        return

    log.info("SWITCH %s: in=%s out=%s", dpid, in_port, out_port)

    msg.actions.append(of.ofp_action_output(port=out_port))
    msg.data = event.ofp

    event.connection.send(msg)


def launch():
    log.info("MyRouter Started")
    core.openflow.addListenerByName("PacketIn", _handle_PacketIn)
```

CODE – myrouter_block.py (Host 3 blocked)

```
from pox.core import core
import pox.openflow.libopenflow_01 as of
from pox.lib.packet.ethernet import ethernet

log = core.getLogger()
```

```
def _handle_PacketIn(event):
    packet = event.parsed

    if not packet.parsed:
        return

    eth = packet.find('ethernet')

    # Ignore IPv6 (reduces errors)
    if eth.type == 0x86DD:
        return

    dpid = event.connection.dpid
    in_port = event.port

    msg = of.ofp_flow_mod()
    msg.match = of.ofp_match.from_packet(packet)

    # STATIC ROUTING + BLOCKING

    # SWITCH 1 (s1)
    if dpid == 1:
        if in_port == 1:
            out_port = 2
        elif in_port == 2:
            out_port = 1
```

```
        else:
            return

    # SWITCH 2 (s2)
    elif dpid == 2:
        if in_port == 2:        # from s1 → to h2
            out_port = 1
        elif in_port == 1:      # from h2 → to s1
            out_port = 2
        elif in_port == 3:      # from s3 → BLOCK
            log.info("Blocked traffic from s3")
            return
        else:
            return

    # SWITCH 3 (s3) → FULL BLOCK
    elif dpid == 3:
        log.info("Blocking all traffic at s3")
        return

    else:
        return
```

```
    log.info("SWITCH %s: in=%s out=%s", dpid, in_port, out_port)

    msg.actions.append(of.ofp_action_output(port=out_port))
    msg.data = event.ofp

    event.connection.send(msg)


def launch():
    log.info("Static Routing (Blocked Version) Started")
    core.openflow.addListenerByName("PacketIn", _handle_PacketIn)
```

6. Test Scenarios

6.1 Normal Scenario (Flooding) – myrouter.py

- All hosts can communicate

Screenshot 2: pingall Output (Normal Scenario)

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
```

All 3 hosts are able to communicate with each other.

6.2 Static Routing (Blocked Scenario)

- h1 ↔ h2 → allowed
- h3 → blocked

Screenshot 3: pingall Output (Blocked Scenario) – myrouter_block.py

```
*** Ping: testing ping reachability
h1 -> h2 X
h2 -> h1 X
h3 -> X X
*** Results: 66% dropped (2/6 received)
mininet>
```

All communications to and from Host 3 blocked. Only h1 to h2 and h2 to h1 works as shown above.

7. Testing and Performance Analysis

7.1 Latency Test (Ping)

Ping is used to measure round-trip delay.

Screenshot 4: Ping Output (Latency Test)

```
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=15.7 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.35 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.086 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.109 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.119 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.083 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.107 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.127 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.133 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.095 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.090 ms
^C
--- 10.0.0.2 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10180ms
rtt min/avg/max/mdev = 0.083/1.634/15.679/4.455 ms
```

7.2 Throughput Test (iperf)

iperf is used to measure bandwidth between hosts.

Screenshot 5: iperf Output (Throughput Test)

```
mininet> iperf
*** Iperf: testing TCP bandwidth between h1 and h3
*** Results: ['47.0 Gbits/sec', '47.0 Gbits/sec']
mininet>
```

7.3 Flow Table Inspection

Flow rules installed by the controller are verified using: dpctl dump-flows

Screenshot 6: Flow Table Entries

```
mininet> dpctl dump-flows
*** s1 ------------------------------------------------------------------------
 cookie=0x0, duration=174.598s, table=0, n_packets=3, n_bytes=270, ipv6,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=33:33:00:00:00:16 actions=output:"s1-eth2"
 cookie=0x0, duration=174.373s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=33:33:ff:5a:62:76 actions=output:"s1-eth1"
 cookie=0x0, duration=174.341s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=33:33:ff:da:94:5a actions=output:"s1-eth1"
 cookie=0x0, duration=174.246s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=33:33:ff:11:f5:4d actions=output:"s1-eth2"
 cookie=0x0, duration=174.054s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=12:98:75:27:b2:d5,dl_dst=33:33:ff:27:b2:d5 actions=output:"s1-eth1"
 cookie=0x0, duration=174.023s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=d6:8d:05:f9:ad:e9,dl_dst=33:33:ff:f9:ad:e9 actions=output:"s1-eth1"
 cookie=0x0, duration=173.958s, table=0, n_packets=3, n_bytes=270, ipv6,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=33:33:00:00:00:16 actions=output:"s1-eth1"
 cookie=0x0, duration=173.893s, table=0, n_packets=5, n_bytes=470, ipv6,vlan_tci=0x0000,dl_src=12:98:75:27:b2:d5,dl_dst=33:33:00:00:00:16 actions=output:"s1-eth1"
 cookie=0x0, duration=173.798s, table=0, n_packets=5, n_bytes=470, ipv6,vlan_tci=0x0000,dl_src=d6:8d:05:f9:ad:e9,dl_dst=33:33:00:00:00:16 actions=output:"s1-eth1"
 cookie=0x0, duration=173.339s, table=0, n_packets=2, n_bytes=180, ipv6,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=33:33:00:00:00:16 actions=output:"s1-eth1"
 cookie=0x0, duration=173.335s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=33:33:00:00:00:02 actions=output:"s1-eth1"
 cookie=0x0, duration=173.310s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=33:33:00:00:00:02 actions=output:"s1-eth1"
 cookie=0x0, duration=173.220s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=33:33:00:00:00:02 actions=output:"s1-eth2"
 cookie=0x0, duration=173.021s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=12:98:75:27:b2:d5,dl_dst=33:33:00:00:00:02 actions=output:"s1-eth1"
 cookie=0x0, duration=172.993s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=d6:8d:05:f9:ad:e9,dl_dst=33:33:00:00:00:02 actions=output:"s1-eth1"
 cookie=0x0, duration=172.938s, table=0, n_packets=14, n_bytes=2260, ipv6,vlan_tci=0x0000,dl_src=12:98:75:27:b2:d5,dl_dst=33:33:00:00:00:fb actions=output:"s1-eth1"
 cookie=0x0, duration=172.910s, table=0, n_packets=14, n_bytes=2260, ipv6,vlan_tci=0x0000,dl_src=d6:8d:05:f9:ad:e9,dl_dst=33:33:00:00:00:fb actions=output:"s1-eth1"
 cookie=0x0, duration=165.610s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,a
rp_op=1 actions=output:"s1-eth2"
 cookie=0x0, duration=165.603s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=8a:fa:6e:11:f5:4d,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,a
rp_op=2 actions=output:"s1-eth1"
 cookie=0x0, duration=160.513s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=8a:fa:6e:11:f5:4d,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,a
rp_op=1 actions=output:"s1-eth1"
 cookie=0x0, duration=160.509s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=2e:97:01:da:94:5a,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,a
rp_op=2 actions=output:"s1-eth2"
 cookie=0x0, duration=76.515s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,ar
p_op=1 actions=output:"s1-eth2"
 cookie=0x0, duration=76.505s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=8a:fa:6e:11:f5:4d,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,ar
p_op=2 actions=output:"s1-eth1"
 cookie=0x0, duration=71.426s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=8a:fa:6e:11:f5:4d,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,ar
p_op=1 actions=output:"s1-eth1"
 cookie=0x0, duration=71.424s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=82:cb:e5:5a:62:76,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,ar
p_op=2 actions=output:"s1-eth2"
 cookie=0x0, duration=165.601s, table=0, n_packets=11, n_bytes=1078, icmp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=2e:97:01:da:94:5a,nw_src=10.0.0.1,nw_dst=10.0.0.2
,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth2"
 cookie=0x0, duration=165.596s, table=0, n_packets=11, n_bytes=1078, icmp,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=8a:fa:6e:11:f5:4d,nw_src=10.0.0.2,nw_dst=10.0.0.1
,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s1-eth1"
```

```
*** s2 ------------------------------------------------------------------------
 cookie=0x0, duration=174.610s, table=0, n_packets=3, n_bytes=270, ipv6,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=33:33:00:00:00:16 actions=FLOOD
 cookie=0x0, duration=174.547s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=2a:80:9c:c1:27:a5,dl_dst=33:33:ff:c1:27:a5 actions=FLOOD
 cookie=0x0, duration=174.386s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=33:33:ff:5a:62:76 actions=FLOOD
 cookie=0x0, duration=174.354s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=33:33:ff:da:94:5a actions=FLOOD
 cookie=0x0, duration=174.259s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=33:33:ff:11:f5:4d actions=FLOOD
 cookie=0x0, duration=174.067s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=12:98:75:27:b2:d5,dl_dst=33:33:ff:27:b2:d5 actions=FLOOD
 cookie=0x0, duration=173.972s, table=0, n_packets=3, n_bytes=270, ipv6,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=33:33:00:00:00:16 actions=FLOOD
 cookie=0x0, duration=173.907s, table=0, n_packets=5, n_bytes=470, ipv6,vlan_tci=0x0000,dl_src=12:98:75:27:b2:d5,dl_dst=33:33:00:00:00:16 actions=FLOOD
 cookie=0x0, duration=173.513s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=2a:80:9c:c1:27:a5,dl_dst=33:33:00:00:00:02 actions=FLOOD
 cookie=0x0, duration=173.510s, table=0, n_packets=3, n_bytes=290, ipv6,vlan_tci=0x0000,dl_src=2a:80:9c:c1:27:a5,dl_dst=33:33:00:00:00:16 actions=FLOOD
 cookie=0x0, duration=173.438s, table=0, n_packets=14, n_bytes=2260, ipv6,vlan_tci=0x0000,dl_src=2a:80:9c:c1:27:a5,dl_dst=33:33:00:00:00:fb actions=FLOOD
 cookie=0x0, duration=173.354s, table=0, n_packets=2, n_bytes=180, ipv6,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=33:33:00:00:00:16 actions=FLOOD
 cookie=0x0, duration=173.353s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=33:33:00:00:00:02 actions=FLOOD
 cookie=0x0, duration=173.330s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=33:33:00:00:00:02 actions=FLOOD
 cookie=0x0, duration=173.231s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=33:33:00:00:00:02 actions=FLOOD
 cookie=0x0, duration=173.038s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=12:98:75:27:b2:d5,dl_dst=33:33:00:00:00:02 actions=FLOOD
 cookie=0x0, duration=172.955s, table=0, n_packets=14, n_bytes=2260, ipv6,vlan_tci=0x0000,dl_src=12:98:75:27:b2:d5,dl_dst=33:33:00:00:00:fb actions=FLOOD
 cookie=0x0, duration=165.621s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,a
rp_op=1 actions=FLOOD
 cookie=0x0, duration=165.618s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=8a:fa:6e:11:f5:4d,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,a
rp_op=2 actions=FLOOD
 cookie=0x0, duration=160.530s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=8a:fa:6e:11:f5:4d,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,a
rp_op=1 actions=FLOOD
 cookie=0x0, duration=160.520s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=2e:97:01:da:94:5a,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,a
rp_op=2 actions=FLOOD
 cookie=0x0, duration=76.525s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,ar
p_op=1 actions=FLOOD
 cookie=0x0, duration=76.519s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=8a:fa:6e:11:f5:4d,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,ar
p_op=2 actions=FLOOD
 cookie=0x0, duration=71.440s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=8a:fa:6e:11:f5:4d,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,ar
p_op=1 actions=FLOOD
 cookie=0x0, duration=71.436s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=82:cb:e5:5a:62:76,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,ar
p_op=2 actions=FLOOD
 cookie=0x0, duration=165.613s, table=0, n_packets=11, n_bytes=1078, icmp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=2e:97:01:da:94:5a,nw_src=10.0.0.1,nw_dst=10.0.0.2
,nw_tos=0,icmp_type=8,icmp_code=0 actions=FLOOD
 cookie=0x0, duration=165.612s, table=0, n_packets=11, n_bytes=1078, icmp,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=8a:fa:6e:11:f5:4d,nw_src=10.0.0.2,nw_dst=10.0.0.1
,nw_tos=0,icmp_type=0,icmp_code=0 actions=FLOOD
```

```
.0.0.1,nw_tos=0,tp_src=5001,tp_dst=59914 actions=FLOOD
*** s3 ------------------------------------------------------------------------
 cookie=0x0, duration=174.616s, table=0, n_packets=3, n_bytes=270, ipv6,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=33:33:00:00:00:16 actions=output:"s3-eth1"
 cookie=0x0, duration=174.547s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=2a:80:9c:c1:27:a5,dl_dst=33:33:ff:c1:27:a5 actions=output:"s3-eth1"
 cookie=0x0, duration=174.397s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=33:33:ff:5a:62:76 actions=output:"s3-eth2"
 cookie=0x0, duration=174.363s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=33:33:ff:da:94:5a actions=output:"s3-eth1"
 cookie=0x0, duration=174.300s, table=0, n_packets=5, n_bytes=470, ipv6,vlan_tci=0x0000,dl_src=56:9a:fd:21:ce:38,dl_dst=33:33:00:00:00:16 actions=output:"s3-eth1"
 cookie=0x0, duration=174.268s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=33:33:ff:11:f5:4d actions=output:"s3-eth1"
 cookie=0x0, duration=173.980s, table=0, n_packets=3, n_bytes=270, ipv6,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=33:33:00:00:00:16 actions=output:"s3-eth1"
 cookie=0x0, duration=173.820s, table=0, n_packets=1, n_bytes=86, ipv6,vlan_tci=0x0000,dl_src=56:9a:fd:21:ce:38,dl_dst=33:33:ff:21:ce:38 actions=output:"s3-eth1"
 cookie=0x0, duration=173.520s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=2a:80:9c:c1:27:a5,dl_dst=33:33:00:00:00:02 actions=output:"s3-eth1"
 cookie=0x0, duration=173.515s, table=0, n_packets=3, n_bytes=290, ipv6,vlan_tci=0x0000,dl_src=2a:80:9c:c1:27:a5,dl_dst=33:33:00:00:00:16 actions=output:"s3-eth1"
 cookie=0x0, duration=173.447s, table=0, n_packets=14, n_bytes=2260, ipv6,vlan_tci=0x0000,dl_src=2a:80:9c:c1:27:a5,dl_dst=33:33:00:00:00:fb actions=output:"s3-eth1"
 cookie=0x0, duration=173.368s, table=0, n_packets=2, n_bytes=180, ipv6,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=33:33:00:00:00:16 actions=output:"s3-eth2"
 cookie=0x0, duration=173.368s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=33:33:00:00:00:02 actions=output:"s3-eth2"
 cookie=0x0, duration=173.332s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=33:33:00:00:00:02 actions=output:"s3-eth1"
 cookie=0x0, duration=173.239s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=33:33:00:00:00:02 actions=output:"s3-eth1"
 cookie=0x0, duration=172.791s, table=0, n_packets=6, n_bytes=420, ipv6,vlan_tci=0x0000,dl_src=56:9a:fd:21:ce:38,dl_dst=33:33:00:00:00:02 actions=output:"s3-eth1"
 cookie=0x0, duration=172.711s, table=0, n_packets=14, n_bytes=2260, ipv6,vlan_tci=0x0000,dl_src=56:9a:fd:21:ce:38,dl_dst=33:33:00:00:00:fb actions=output:"s3-eth1"
 cookie=0x0, duration=165.628s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,a
rp_op=1 actions=output:"s3-eth1"
 cookie=0x0, duration=165.625s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=8a:fa:6e:11:f5:4d,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,a
rp_op=2 actions=output:"s3-eth1"
 cookie=0x0, duration=160.534s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=8a:fa:6e:11:f5:4d,arp_spa=10.0.0.2,arp_tpa=10.0.0.1,a
rp_op=1 actions=output:"s3-eth1"
 cookie=0x0, duration=160.528s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=2e:97:01:da:94:5a,arp_spa=10.0.0.1,arp_tpa=10.0.0.2,a
rp_op=2 actions=output:"s3-eth1"
 cookie=0x0, duration=76.534s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,ar
p_op=1 actions=output:"s3-eth1"
 cookie=0x0, duration=76.532s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=8a:fa:6e:11:f5:4d,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,ar
p_op=2 actions=output:"s3-eth2"
 cookie=0x0, duration=71.452s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=8a:fa:6e:11:f5:4d,arp_spa=10.0.0.3,arp_tpa=10.0.0.1,ar
p_op=1 actions=output:"s3-eth2"
 cookie=0x0, duration=71.445s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=82:cb:e5:5a:62:76,arp_spa=10.0.0.1,arp_tpa=10.0.0.3,ar
p_op=2 actions=output:"s3-eth1"
 cookie=0x0, duration=165.619s, table=0, n_packets=11, n_bytes=1078, icmp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=2e:97:01:da:94:5a,nw_src=10.0.0.1,nw_dst=10.0.0.2
,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s3-eth1"
 cookie=0x0, duration=165.616s, table=0, n_packets=11, n_bytes=1078, icmp,vlan_tci=0x0000,dl_src=2e:97:01:da:94:5a,dl_dst=8a:fa:6e:11:f5:4d,nw_src=10.0.0.2,nw_dst=10.0.0.1
,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s3-eth1"
 cookie=0x0, duration=76.525s, table=0, n_packets=4, n_bytes=272, tcp,vlan_tci=0x0000,dl_src=8a:fa:6e:11:f5:4d,dl_dst=82:cb:e5:5a:62:76,nw_src=10.0.0.1,nw_dst=10.0.0.3,nw_
tos=16,tp_src=59908,tp_dst=5001 actions=output:"s3-eth1"
 cookie=0x0, duration=76.524s, table=0, n_packets=2, n_bytes=140, tcp,vlan_tci=0x0000,dl_src=82:cb:e5:5a:62:76,dl_dst=8a:fa:6e:11:f5:4d,nw_src=10.0.0.3,nw_dst=10.0.0.1,nw_
tos=0,tp_src=5001,tp_dst=59908 actions=output:"s3-eth2"
```

8. Validation (Regression Testing)

Flow rules are deleted and reinstalled to verify consistency.

Screenshot 7: Regression Test (After Deleting Flows)

```
mininet> sh ovs-ofctl del-flows s1
mininet> sh ovs-ofctl del-flows s2
mininet> sh ovs-ofctl del-flows s3
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet>
```

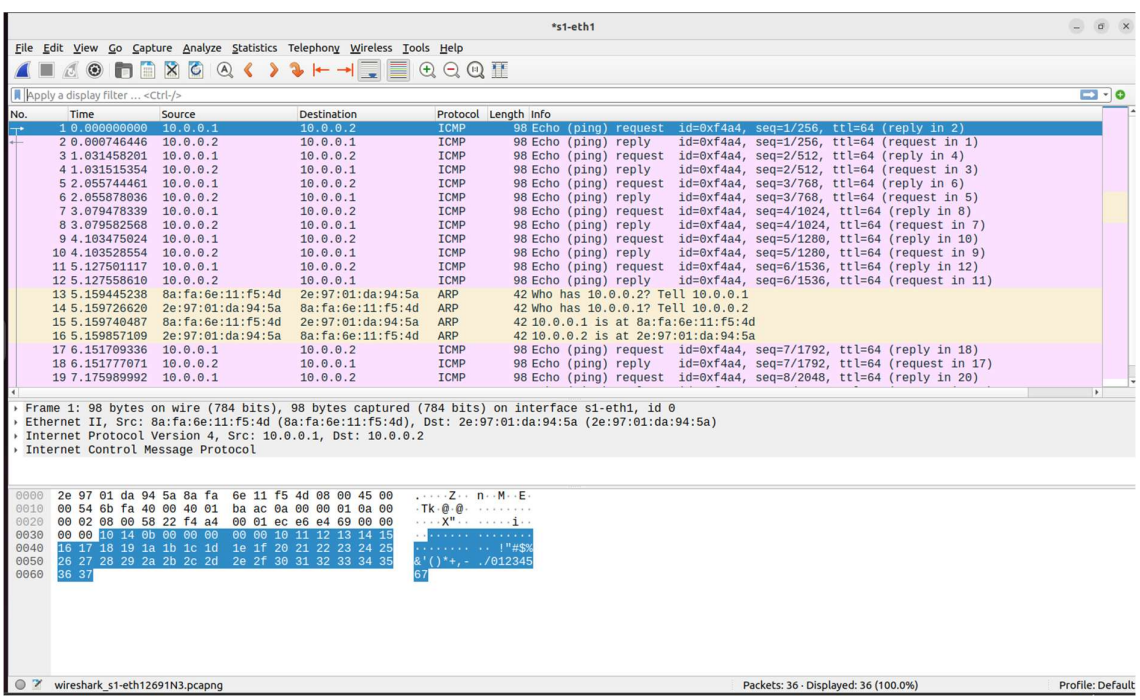
Flow rules reinstalled automatically during pingall even though del-flows was used to delete the flow rules.

9. Packet Analysis (Wireshark)

Wireshark is used to observe:

- ARP packets (IP to MAC resolution)
- ICMP packets (ping request/reply)

Screenshot 8: Wireshark Capture (ARP & ICMP)



10. Results

- Successfully implemented static routing using an SDN controller (POX) in a Mininet environment
- Verified correct installation and execution of match–action flow rules using OpenFlow
- Demonstrated two network scenarios:
    - Full connectivity using flooding
    - Controlled communication with traffic blocking (h3 restricted)
- Observed proper controller–switch interaction through PacketIn events and flow rule installation
- Validated network behavior using flow tables and controller logs
- Measured latency using ping (ICMP) and throughput using iperf, confirming efficient data transmission

11. Conclusion

This project demonstrates the implementation of static routing in a Software Defined Network (SDN) using a POX controller. By separating the control plane from the data plane, the controller was able to centrally manage network behavior through programmable flow rules.

The use of match–action logic allowed precise control over packet forwarding, enabling both full connectivity and restricted communication scenarios. The results highlight how SDN simplifies network management, improves flexibility, and allows dynamic enforcement of network policies such as traffic blocking.

Overall, the project validates the effectiveness of SDN in providing centralized, scalable, and programmable network control, making it a powerful approach for modern networking environments.